Geometric Deep Learning for Molecular Dynamics: A Framework for Protein–Ligand Binding Affinity and Conformational Dynamics Prediction

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***Abstract*—Computational drug discovery relies on accurate modeling of protein–ligand binding, yet classical molecular dynamics (MD) simulations are prohibitively expensive, and most geometric deep learning (GDL) models predict only a static binding pose while ignoring conformational dynamics and quantum-mechanical effects. We present GraphMD, a multiscale geometric deep learning framework that couples a residue-level protein graph with an atom-level binding-pocket graph through cross-scale attention, augments the protein graph with MD-derived dynamic correlation edges, and employs a stability-aware two-stage training strategy (frame-level stability pretraining followed by binding-affinity fine-tuning) to overcome limited labeled affinity data. We further compare two message-passing schedules for the dual-graph architecture — a *sequential* variant (M1) and an *interleaved* variant (M2) — for the stability pretraining task. On the CASF-2016 benchmark, the fine-tuned GraphMD model achieves a binding affinity RMSE of 1.214, outperforming FAST, IGN, OnionNet, SIGN, GIGN and CurvAGN, while the pretrained encoder achieves a stability (RMSD) RMSE of 0.903 on CASF-2016. These results indicate that combining static geometry with MD-informed dynamics and staged training improves prediction accuracy over static-only baselines.**

***Index Terms*—protein–ligand binding affinity, geometric deep learning, graph neural networks, molecular dynamics, drug discovery**

I. Introduction

Computational drug discovery relies heavily on understanding how proteins and ligands bind together. Conventional docking algorithms typically treat the receptor as rigid and only the ligand as flexible, which is computationally efficient but produces incorrect poses whenever induced-fit binding occurs, whereas MD simulation removes this assumption by numerically integrating the equations of motion for every atom at a steep computational cost [1]. Geometric deep learning instead learns directly from 3D atomic structure and geometry, enabling much faster predictions [2]. Recent equivariant generative models such as DynamicBind have shown that ligand-specific protein–ligand complex structures can be predicted directly from learned representations rather than repeated simulation [14]. However, most existing GDL models still predict only a static pose or a single equilibrium structure, missing the temporal dynamics and electronic effects that determine true binding strength. This project introduces GraphMD, a multiscale graph neural network (GNN) framework that jointly learns conformational stability and binding affinity, bridging the gap between fast static scoring functions and expensive, dynamics-aware MD simulation.

Drug discovery proceeds through target identification, hit identification (virtual/high-throughput screening), lead optimization, preclinical studies and clinical trials. Within hit identification, AI/ML models predict binding energy and structural stability (RMSD) to prioritize candidates, reducing wasted experimental resources. Two quantities are central to this process: *binding affinity*, which measures how strongly a ligand binds its target, and *RMSD*, which tracks how much a ligand's atomic positions deviate over an MD trajectory and therefore how stable the binding pose is.

Motivation. Drug discovery is slow and costly, making efficient computational alternatives essential. MD simulations are accurate but rely on force-field approximations and heavy computation, while current ML scoring models often assume rigid proteins and static classical descriptors, missing flexibility, induced-fit effects, and quantum-mechanical contributions to binding.

Problem statement. Current computational models largely treat molecular binding as static, oversimplifying temporal dynamics and neglecting electronic effects. The core challenge addressed in this work is the joint modeling of binding energetics and conformational dynamics through improved temporal encoding, since existing deep learning methods that assume rigid, single-frame proteins limit generalization and accuracy.

Research objectives. This work aims to: (1) develop robust SE(3)-aware geometric representations for protein–ligand complexes; (2) create encoding strategies for incorporating MD trajectory information into a graph-based model through dynamic correlation edges; (3) design a two-stage, stability-aware training framework that can predict both binding energetics and conformational stability; and (4) evaluate the framework, together with an ablation on message-passing schedules, against state-of-the-art static-structure baselines.

II. Related Work

Prior to the rise of geometric deep learning, protein–ligand scoring relied on physics-based force fields, empirical regression against hand-crafted contact descriptors, or knowledge-

based statistical potentials [2]; these approaches share the limitation that they sum pairwise interaction terms additively and treat solvation and entropy only approximately. Early machine-learning scorers such as RF-Score replaced hand-tuned energy terms with a random forest trained on a 36-dimensional vector of atom-pair contact counts, improving ranking accuracy but still depending heavily on feature engineering and offering limited transferability across protein families [3]. Large curated resources such as PDBbind subsequently provided the labeled structure–affinity pairs needed to train and benchmark such models [4]. The introduction of graph neural networks addressed the feature-engineering bottleneck by learning interaction features directly from atomic coordinates and connectivity, while equivariant architectures such as SchNet's continuous-filter convolutions further ensured that predictions do not depend on the arbitrary orientation of the input structure [6]. GraphMD builds on this line of graph-based, geometry-aware scoring and extends it with explicit dynamical and multiscale components.

Multi-task and pose-evaluation GNNs. IGModel is a geometric GNN with rotational and translational equivariance that jointly predicts RMSD (pose accuracy) and pK_d (binding affinity) from a shared representation using two prediction heads, exploiting the correlation between structural accuracy and binding thermodynamics [7]. GNN-DOVE evolves CNN-based docking-pose evaluation into a graph-based formulation, representing extracted protein–protein interface regions as graphs of atomic nodes and distance-based edges and using attention to highlight critical interaction regions [5]. Both models demonstrate that geometric graph representations outperform earlier voxel- or descriptor-based scoring, but their interaction representations remain built from a single structure and do not explicitly encode trajectory-level dynamics.

Equivariant pose and docking prediction. EquiDock reformulates rigid protein–protein docking as SE(3)-equivariant keypoint matching, predicting a rigid-body transformation that aligns two protein graphs without explicit conformational sampling [8]. EQuiBind extends this idea to small-molecule docking, directly positioning a ligand within a protein pocket in a single shot rather than through iterative energy-based search [9]. DiffDock instead frames docking as conditional generative modeling, learning a diffusion process over the physically meaningful translation, rotation and torsional degrees of freedom of the ligand and using a learned confidence model to rank sampled poses [10]. These models substantially reduce the computational cost of pose generation but, like IGModel and GNN-DOVE, are not designed to predict binding affinity or dynamic stability from MD trajectories.

Physics-informed and dynamics-aware models. ECBind tokenizes electron-cloud density around the binding site using a structure-aware transformer with hierarchical codebooks, enabling binding modes influenced by polarization and charge transfer to be represented beyond what atom-level models capture [12]. TANKBind incorporates trigonometry-aware neural networks to improve geometric consistency in drug–protein binding structure prediction [13]. Boltz-2 is a foundation model for ensemble-aware affinity prediction that aggregates signals from multiple protein conformations drawn from simulations and experimentally resolved structures, but it ultimately produces predictions from aggregated ensemble representations rather than from time-resolved trajectories [15].

From this literature we identify recurring design principles: (i) multi-resolution or multi-task representations of protein–ligand systems outperform single static structures; (ii) incorporating flexibility or ensemble information improves affinity prediction; and (iii) there remains a persistent gap between models that reason about dynamics implicitly (through ensembles or electron-density snapshots) and models that learn directly from time-resolved, frame-by-frame MD trajectories. This gap — together with the general omission of quantum-mechanical effects from purely geometric scoring functions — motivates GraphMD.

III. Methodology

A. Datasets

GraphMD is trained on two dataset families. MISATO provides curated MD trajectories of approximately 100 frames per complex for protein–ligand systems [11], while PDBbind supplies experimentally measured binding affinities for a large set of resolved complexes [4]. PDB identifiers from MISATO complexes are matched against PDBbind to retrieve experimental K_d/K_i/IC_{50} values, which are converted to a single $\mathrm{pK}_d = -\log_{10}(K_d \text{ in mol/L})$ target, producing a paired (structure, trajectory, affinity) dataset. The resulting data is split 70:15:15 into 11,872 training, 2,550 validation and 2,550 test complexes. CASF-2016 (285 complexes), disjoint from the MISATO split, is used as an external benchmark for both stability and affinity prediction.

B. Dual-Graph Architecture

GraphMD represents each protein–ligand complex with two coupled graphs: a residue-level protein graph and an atom-level binding-pocket graph, linked by cross-scale attention (Fig. 1).

Residue-level protein graph (global context). Nodes are protein residues (≈300 per complex), each carrying a 20-dimensional one-hot residue-identity encoding and a local backbone frame built from the N, C_α and C atoms,

$$F_{\text{res}} = \big(\hat{u}(N-C_\alpha),\ \hat{u}(C_\alpha-C),\ \hat{u}(N-C_\alpha)\times\hat{u}(C_\alpha-C)\big), \tag{1}$$

which is invariant to the arbitrary placement of the structure in space. Edges are constructed from two complementary sources: sequential backbone edges connecting each residue to its immediate chain neighbors, and spatial cutoff edges connecting any pair of residues whose C_α atoms lie within 8 Å. In addition, *dynamic correlation edges* are derived from a Dynamic Cross-Correlation Matrix (DCCM) computed from the C_α displacement vectors observed across 20 sampled MD frames: for each residue pair, the Pearson

correlation of their fluctuation vectors over time is computed, and only the Top-K most strongly correlated neighbors per residue are retained as a distinct edge type, preventing the otherwise dense N^2 correlation matrix from causing an edge explosion. A relational graph attention network with separate weight matrices per edge type then learns structural and dynamic relationships simultaneously.

Atom-level pocket graph (local interactions). Nodes are all ligand atoms together with protein atoms located within an 8–10 Å radius of the ligand in a single fixed reference frame, so that the pocket topology does not change from frame to frame. Each atom node carries a one-hot element encoding, hybridization state, formal charge, and a ligand/protein flag. Beyond categorical features, an SE(3)-equivariant pair frame is built directly from relative atomic positions,

$$F_{\text{atom}} = \big(\hat{u}(x_i - x_j),\ \hat{u}(x_i \times x_j),\ \hat{u}(x_i - x_j) \times \hat{u}(x_i \times x_j)\big), \tag{2}$$

so that rotating or translating the complex does not change the resulting features. Distance-cutoff edges connect atoms within interaction range, capturing both intramolecular bonds and intermolecular (protein–ligand) contacts.

C. Message Passing and Cross-Scale Attention

Information flows through a three-stage cycle repeated five times. **Stage 1** passes the residue graph through a graph attention layer, propagating information along both edge types so that, over several cycles, even a distal allosteric residue can influence embeddings near the pocket. **Stage 2** performs cross-attention transfer from residues to pocket atoms,

$$h_i^{\text{atom}} \leftarrow h_i^{\text{atom}} + \sum_{j \ni i} \alpha_{ij}\, h_j^{\text{res}}, \quad \alpha_{ij} = \text{softmax}_j\big(\text{attn}(h_i^{\text{atom}}, h_j^{\text{res}})\big), \tag{3}$$

where queries are drawn from atom embeddings and keys/values from residue embeddings, so that global structural context is transferred into local atomic representations without collapsing the two resolutions into one. **Stage 3** runs a single equivariant message-passing layer (128-dimensional hidden state) on the enriched pocket graph, fusing local chemical identity, neighborhood geometry, and the transferred global context; layer normalization is applied after each step.

D. Readout

Atomic embeddings are aggregated into a fixed-size representation via learned attention pooling, applied separately to ligand atoms and protein-pocket atoms to preserve their distinct contributions, and concatenated as $z = [z_{\text{prot}} \| z_{\text{lig}}]$. This vector is passed through a three-layer MLP (256, 128, 64 units, ReLU, dropout 0.2) that outputs a single scalar prediction, trained by minimizing mean squared error against the experimental target.

E. Two-Stage Training Strategy

Given the limited size of labeled affinity data (on the order of 20,000 complexes relative to the parameter count of the architecture), GraphMD exploits the temporal dimension of MD trajectories in two stages (Fig. 2). In **Stage 1 (frame-level stability pretraining)**, 10–20 frames are sampled per MD trajectory, and each frame is labeled with its C_α RMSD relative to the fixed reference structure, with lower RMSD indicating a more stable, lower-energy conformation; the full dual-graph architecture is trained end-to-end to predict this per-frame score using MSE loss, multiplying the effective training signal available from each complex well beyond the single affinity label PDBbind provides. In **Stage 2 (binding-affinity fine-tuning)**, the single sampled frame with the lowest predicted RMSD is selected as the representative structure for that complex, and the pretrained weights are fine-tuned in full (no layers frozen), with the pooling and regression head now trained against the experimental pK_d label using a lower learning rate than pretraining to avoid overwriting the learned structural representations.

F. Model Variants for Stability Pretraining

To determine how information should flow between the residue and pocket graphs during Stage-1 pretraining, we compare two dual-graph message-passing schedules. **M1 (Sequential)** processes the residue graph fully, followed by a single cross-attention transfer, followed by full processing of the pocket graph, i.e., residue-level and atom-level reasoning happen in two separate, non-interacting blocks. **M2 (Interleaved)** instead repeats the residue-graph layer, cross-attention transfer, and pocket-graph layer in sequence across all five cycles, allowing residue and atom representations to refine each other iteratively. Both variants are trained identically on the stability task and evaluated by RMSE against ground-truth frame-level RMSD (Fig. 4). M2 converges to an RMSE of approximately 1.0 with a smoother trajectory across epochs, while M1 stabilizes around an RMSE of approximately 1.1 with visibly higher variance, indicating that repeated, interleaved cross-scale communication is more effective than a single sequential hand-off; M2 is therefore adopted as the message-passing schedule for the final GraphMD pretrained model.

G. Implementation Challenges

Two practical issues arose during graph construction. First, because pocket atoms fluctuate across the ∼100 MD frames of a trajectory, recomputing the 8–10 Å pocket boundary independently for every frame would cause the node set, and hence the graph topology, to change from frame to frame, destroying training stability; this was resolved by anchoring the pocket definition to a single fixed reference structure. Second, because the dynamic correlation matrix is dense by construction, connecting every residue to all correlated partners causes the number of edges to grow combinatorially with protein size; this was resolved with a Top-K correlated-neighbor selection rule, combined with a hard cap on edges

per node, that retains only the strongest, most physically meaningful long-range couplings.

H. Data Efficiency Rationale

Labeled binding-affinity data is scarce relative to typical deep-learning datasets: MISATO and PDBbind together yield on the order of 20,000 experimentally labeled complexes, which is modest for training a multiscale GNN with cross-attention from scratch. MISATO's MD trajectories, in contrast, offer roughly 100 frames per complex that share the same underlying physical and chemical properties that determine affinity, even though individual frames are not directly labeled with affinity values [11]. The two-stage training strategy is designed specifically to exploit this asymmetry: stability pretraining converts each trajectory into 10–20 weakly supervised training examples, multiplying the effective size of the pretraining set by an order of magnitude before the smaller affinity-labeled set is used for fine-tuning.

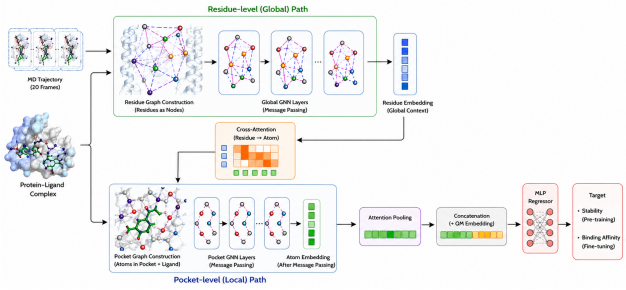


Fig. 1: System Architecture

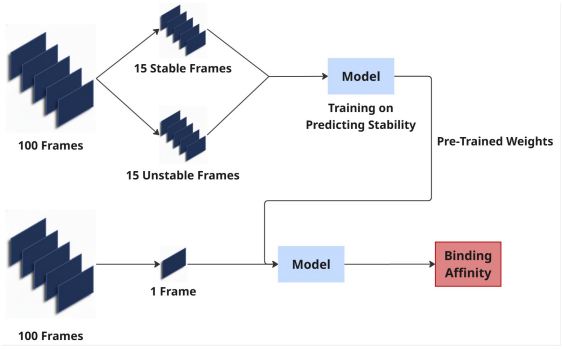


Fig. 2: Pretraining and FIne Tuning process

IV. Experimental Setup

Development and training were performed in Kaggle notebooks with GPU acceleration (NVIDIA T4/P100), using PyTorch and PyTorch Geometric, with NumPy, SciPy and Pandas used for pairwise distance computation, Pearson correlation, and metadata alignment. Because MISATO and its per-complex MD trajectories are too large to load fully into memory, complexes were streamed from read-only storage and graphs were constructed on the fly, with training data processed in mini-batches and periodic checkpoints (model weights, optimizer state, epoch number) saved to allow training to resume under limited memory budgets. Table I summarizes the training configuration.

TABLE I: Experimental setup.

Parameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	8
Epochs	30
Loss function	MSE
Framework	PyTorch / PyTorch Geometric
Hardware	NVIDIA T4 (16 GB)
Platform	Kaggle

V. Results

A. Ablation: Quantum-Mechanical Features Alone

As an ablation, an MLP trained only on quantum-mechanical (QM) descriptors, without any graph architecture, provides a reasonable baseline but its validation performance saturates at an RMSE of approximately 1.45 (Fig. 3), noticeably worse than the full multiscale GraphMD model, indicating that geometric graph structure contributes substantially beyond quantum descriptors alone.

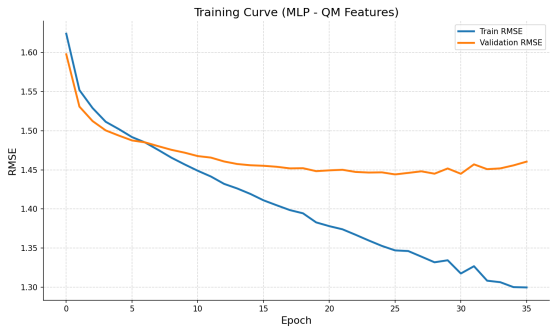


Fig. 3: Train and validation RMSE curves for the MLP model trained using only quantum-mechanical (QM) features

B. Message-Passing Schedule Comparison

Fig. 4 compares the M1 (sequential) and M2 (interleaved) message-passing schedules on the stability-pretraining task. M2 converges faster and more smoothly to an RMSE of approximately 1.0, while M1 oscillates around an RMSE of approximately 1.1, confirming that interleaved cross-scale communication between the residue and pocket graphs is the more effective and stable schedule.

C. Stability and Binding Affinity Prediction

Table II summarizes the performance of the pretrained (stability) and fine-tuned (affinity) GraphMD models on both the held-out MISATO test set and the external CASF-2016 benchmark.

On the CASF-2016 binding-affinity benchmark, GraphMD achieves the lowest RMSE among all compared scoring functions (Table III). Fig. 5 shows predicted-vs-true pK_d scatter

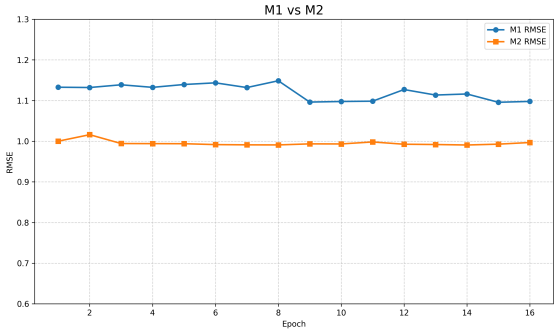


Fig. 4: Comparison of RMSE values for M1 (Sequential Message Passing) and M2 (Interleaved Message Passing) across training epochs for stability prediction.

TABLE II: GraphMD performance across training stages and evaluation sets.

Model	Task	CASF-2016 (285 complexes)	Test set (2550 complexes)
Pretrained	Stability (RMSD)	0.903	1.099
Fine-tuned	Affinity (pK_d)	1.214	1.238

plots alongside the corresponding residual plot; residuals are approximately zero-centered across the prediction range, with no strong systematic bias toward over- or under-prediction.

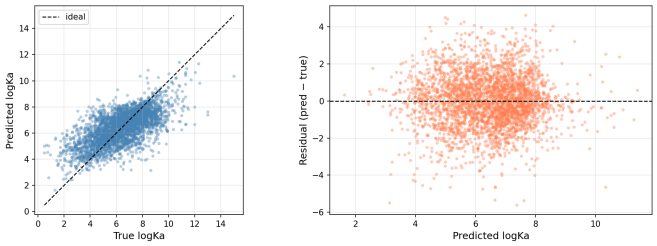


Fig. 5: Scatter plot of true versus predicted binding affinity values (a) alongside the residual plot showing the prediction error, Predicted − True (b).

TABLE III: Binding affinity prediction on CASF-2016 (285 complexes).

Model	RMSE
FAST	1.308
IGN	1.303
OnionNet	1.278
SIGN	1.278
GIGN	1.272
CurvAGN	1.217
GraphMD (Ours)	**1.214**

VI. Discussion

The CASF-2016 results indicate that **multiscale modeling combined with dynamics pays off**: GraphMD attains the lowest affinity error among the compared models, and its pretrained encoder separately gives accurate structural (RMSD) predictions, showing that combining residue-level context, MD-derived dynamics, and stability-focused pretraining improves performance over models that rely only on static geometric features such as FAST, IGN, OnionNet, SIGN, GIGN and CurvAGN. This gain, however, **has a cost**: the improvement depends on having approximately 100 MD frames per complex and a two-stage training pipeline, which makes the method heavier and slower, and limits its use on newly encountered proteins for which simulation data is unavailable. The comparison between M1 and M2 further shows that repeated, interleaved cross-scale communication between graphs is more effective than a single hand-off, suggesting that iterative refinement between resolutions is an important design choice for multiscale molecular GNNs, consistent with the way equivariant architectures such as SchNet and later SE(3)-equivariant docking models [6], [8] have shown that respecting molecular geometry throughout the network, rather than only at the input, improves downstream accuracy.

Limitations. Three limitations follow directly from the current design. First, GraphMD assumes the ligand binding pose is already known, which holds for benchmark datasets such as CASF-2016 but not for fully automated virtual screening, where the pose itself must first be predicted, as in EquiDock, EQuiBind or DiffDock [8]–[10]. Second, the model requires coordinates of trajectory frames per complex to build its dynamic correlation edges and to run stability pretraining; a newly encountered protein without existing simulation data therefore cannot be scored until a trajectory is generated, which limits applicability in rapid, high-throughput screening settings. Third, the multiscale dual-graph architecture combined with two-stage, frame-level pretraining incurs substantially higher preprocessing and training cost than single-graph, static-structure baselines, which constrains rapid experimentation and large-scale hyperparameter search.

VII. Conclusion

GraphMD integrates static structural data with dynamic molecular correlations to improve protein–ligand binding-affinity prediction, addressing the limitations of traditional static models by capturing long-range allosteric effects and conformational flexibility through a multiscale, relational graph neural network with cross-scale attention. On the CASF-2016 benchmark, the fine-tuned model achieves an RMSE of 1.214, outperforming FAST, IGN, OnionNet, SIGN, GIGN and CurvAGN, while the pretrained stability model achieves an RMSD-based RMSE of 0.903, and the interleaved message-passing variant (M2) is shown to be a more effective and stable choice than sequential message passing (M1) for the stability-pretraining stage. The framework provides a scalable foundation for bridging molecular dynamics and geometric deep learning in drug discovery, and establishes a basis for future integration with quantum-mechanical descriptors and generative pose prediction.

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